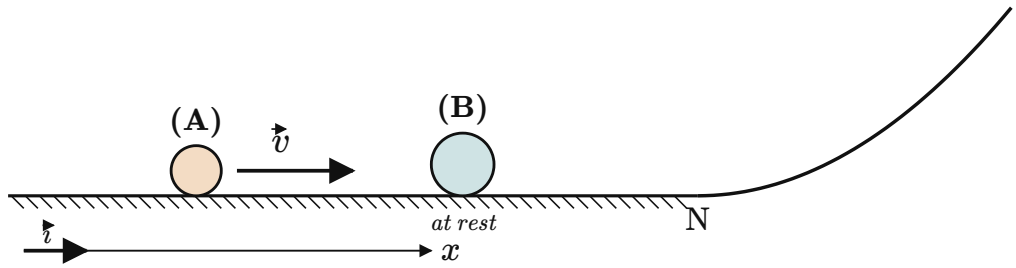
Grade 12 — Life Sciences Lebanese curriculum	**Practice Exam 4** Chapters 1 & 2 — Energy · Linear Momentum	Duration: 2 hours Total: 20 points

This exam is formed of three obligatory exercises. The use of a non-programmable calculator is allowed.

Take, in all exercises: $g = 10\ \text{m/s}^2$.

Exercise 1 — Finding an unknown mass (7 points)

On a horizontal table, a ball (A) of mass $m_A = 200$ g, moving with the velocity $\vec{v}_A = 3\,\vec{i}$ (m/s), collides head-on with a ball (B) of **unknown mass** m_B, initially at rest (Doc. 1). The collision is **perfectly elastic**. Just after the collision, (A) rebounds with the velocity $\vec{v}'_A = -1\,\vec{i}$ (m/s) and (B) moves with the velocity $\vec{v}'_B = v'_B\,\vec{i}$. Friction is neglected; the balls are considered as particles.



Doc. 1

Part 1 — Before the collision

1.1 Show that the system [(A), (B)] is pseudo-isolated.

1.2 Calculate the linear momentum of the system before the collision.

Part 2 — The two unknowns

2.1 Write the two equations that express the conservations during this collision, in terms of m_A, m_B, v_A, v'_A and v'_B.

2.2 Determine v'_B and the mass m_B.

Part 3 — After the collision

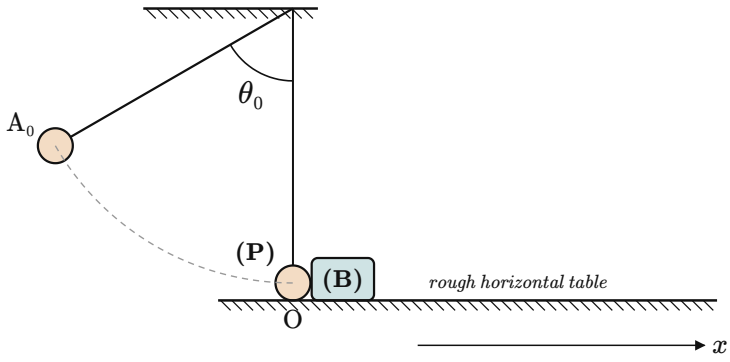
(B) reaches the point N, then climbs a frictionless ramp. The horizontal plane of the table is the reference level of gravitational potential energy.

3.1 Calculate the maximum height h reached by (B) on the ramp.

3.2 Calculate the variation $\Delta\vec{P}_A$ of the linear momentum of (A) during the collision. Deduce the average force exerted by (B) on (A), knowing that the contact lasts 10 ms.

Exercise 2 — A pendulum hits a block (7 points)

A pendulum (P) is formed of a light inextensible string of length $L = 0.9$ m and a small ball of mass $m = 200$ g. It is released without initial velocity from A_0, the string making the angle $\theta_0 = 60°$ with the vertical (Doc. 2). At its lowest position O, the ball hits a block (B), of the **same mass** m, at rest on a horizontal table. Air resistance is neglected.



Doc. 2

> The horizontal plane through O is the reference level of gravitational potential energy.

Part 1 — The descent

1.1 Calculate the mechanical energy of the system [(P), Earth] at A_0.

1.2 Deduce the speed v_1 of the ball just before hitting (B).

Part 2 — The elastic collision

The collision is head-on and **perfectly elastic**. Let v_1' and v_2' be the algebraic values of the velocities of the ball and of (B) just after the collision.

2.1 Write the two equations that express the conservations during the collision.

2.2 Show that the ball stops and that (B) moves with the speed $v_2' = v_1$.

Part 3 — (B) on the rough table

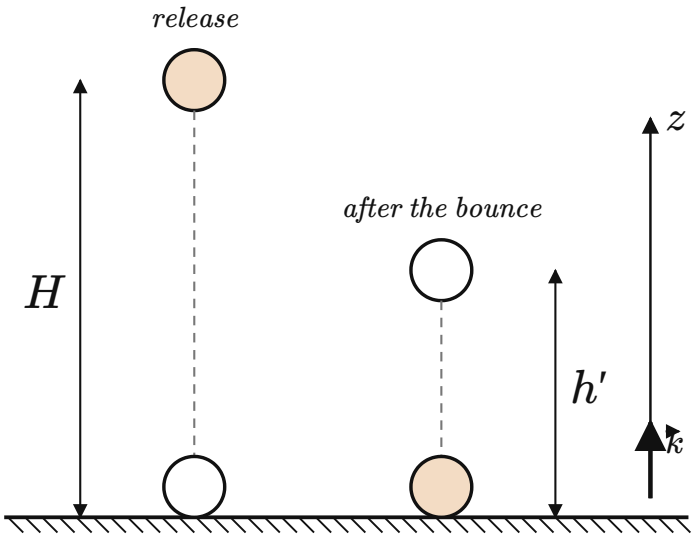
(B) slides on the table, acted upon by a constant friction force $\vec{f}$, and stops after covering the distance $d = 1.5$ m.

3.1 Calculate the variation of the mechanical energy of the system [(B), Earth]. Deduce the magnitude f.

3.2 Using Newton's second law $\Sigma\vec{F} = \dfrac{d\vec{P}}{dt}$, calculate the duration of the motion of (B) on the table.

Exercise 3 — A bouncing ball (6 points)

A ball of mass $m = 0.5$ kg is released without initial velocity from a height $H = 1.8$ m above the ground. It bounces on the ground and rises vertically. The vertical axis $(O, \vec{k})$ is directed upwards, O on the ground. The ground is the reference level of gravitational potential energy. Air resistance is neglected.



Doc. 3

Part 1 — The fall

1.1 Using the conservation of mechanical energy, calculate the speed v_1 of the ball just before it hits the ground.

Part 2 — The bounce

The ball leaves the ground with the speed $v_2 = 4$ m/s. The contact with the ground lasts $\Delta t = 0.05$ s.

2.1 Calculate the variation $\Delta\vec{P}$ of the linear momentum of the ball during the contact.

2.2 Deduce the average force exerted by the ground on the ball.

Part 3 — After the bounce

3.1 Calculate the kinetic energy lost during the bounce. Is the collision with the ground elastic? Justify.

3.2 Calculate the maximum height h' reached by the ball after the bounce.

— End of the exam —